Monopolistic Competition

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 17, Monopolistic Competition.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 17, "Monopolistic Competition".

What you must know from this chapter

1. Definition: many firms, differentiated products, free entry.
2. Why each firm faces a downward-sloping demand curve.
3. The short run: MR $=$ MC, the price from demand, profit or loss on the graph.
4. The long run: entry and exit drive profit to zero, $P =$ ATC.
5. Comparison with perfect competition: excess capacity and a markup, $P >$ MC.

Part 1

What Is Monopolistic Competition?

Monopolistic competition

Definition: Monopolistic competition

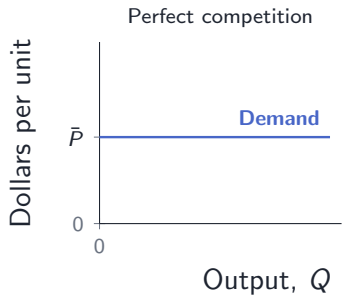
A market with many firms selling products that are similar but not identical, and free entry and exit.

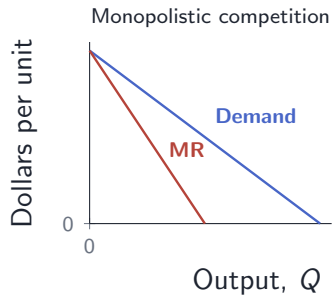
- **Many sellers:** many firms compete for the same group of buyers.
- **Product differentiation:** each firm's product is slightly different from its rivals'.
- **Free entry and exit:** firms can enter or leave the market without restriction.

Examples: restaurants, hair salons, coffee shops, clothing brands.

Each firm faces a downward-sloping demand curve

- A restaurant that raises its price loses some customers to rivals.
- It does not lose them all: some prefer its food.
- So its demand curve slopes down.
- As for a monopoly, $P > \text{MR}$.





Between perfect competition and monopoly

	Perfect competition	Monopolistic competition	Monopoly
Number of firms	Many	Many	One
Product	Identical	Differentiated	No close substitutes
Entry	Free	Free	Blocked
Firm's demand	Horizontal	Slopes down	Slopes down

Finding

Like perfect competition: many firms and free entry. Like monopoly: a downward-sloping demand curve, so $P >$ MR.

Part 1 summary

- Monopolistic competition: many firms, differentiated products, free entry and exit.
- Each firm's product differs a little from its rivals'.
- So each firm faces a downward-sloping demand curve.
- So, as for a monopoly, $P >$ MR.
- Compared with perfect competition: the same many firms and free entry, but differentiated products.

Your turn: Check your understanding

1. Which market is monopolistically competitive?
   A. Wheat farming. B. Hair salons in a city. C. A town's water network.
2. A hair salon raises its price. It loses:
   A. none of its customers. B. some of its customers. C. all of its customers.
3. New salons can open whenever they like.
   A. True B. False

Solution: Check your understanding

1. **B.** Many salons, each slightly different, free entry. Wheat is identical (perfect competition); the water network is a natural monopoly.
2. **B.** Some customers prefer this salon, so not all leave. Demand slopes down.
3. **A.** True: entry is free.

Check: Part 1

A monopolistically competitive industry is characterized by

A many firms, differentiated products, and barriers to entry.

B many firms, differentiated products, and free entry.

C a few firms, identical products, and free entry.

D a few firms, differentiated products, and barriers to entry.

Check: Part 1 – answer

A monopolistically competitive industry is characterized by

A many firms, differentiated products, and barriers to entry.

B many firms, differentiated products, and free entry.

C a few firms, identical products, and free entry.

D a few firms, differentiated products, and barriers to entry.

Finding

The three features of monopolistic competition: many firms, differentiated products and free entry.

Part 2

The Short Run: Act Like a Monopoly

The same three steps as a monopoly

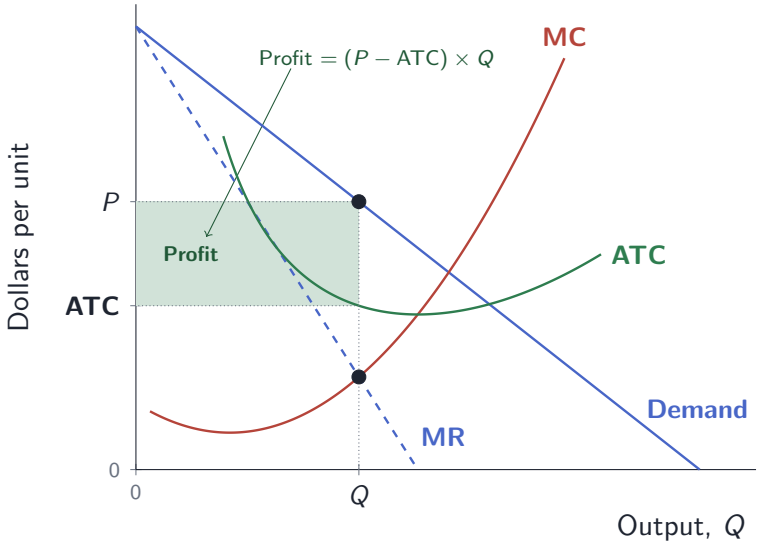
1. **Output:** produce Q where MR $=$ MC.
2. **Price:** go up from Q to the demand curve. That is the price, P.
3. **Profit:** compare P with ATC at Q:

$$\text{Profit} = (P - \text{ATC}) \times Q.$$

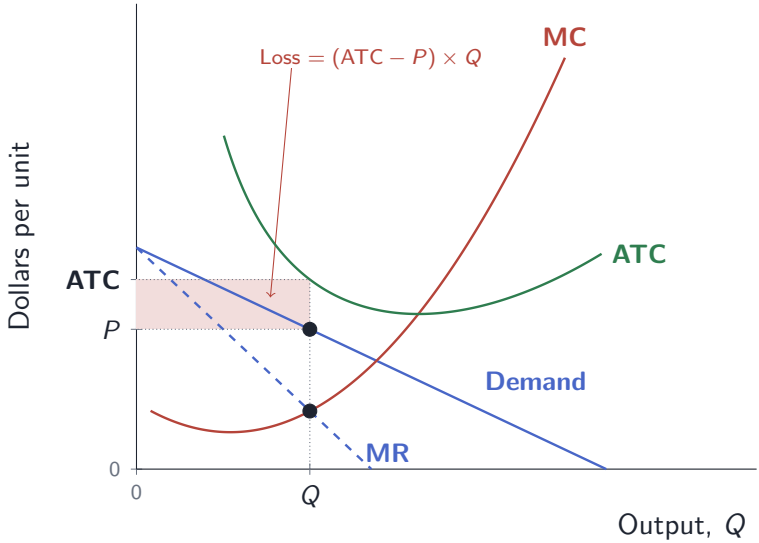
Finding

If $P >$ ATC, the firm makes a profit. If $P <$ ATC, it makes a loss.

Short-run profit

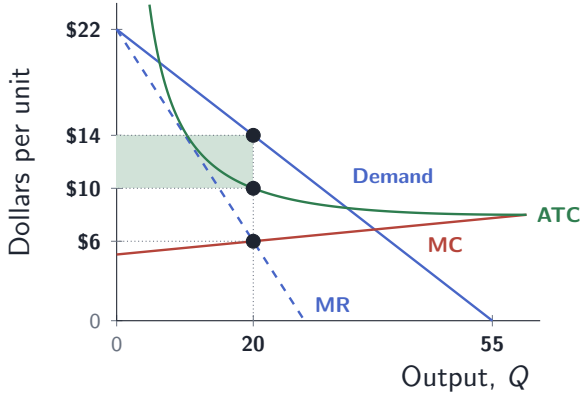


Short-run loss



Example: reading profit from the graph

1. MR = MC at $Q = 20$.
2. Up to demand: $P = \$14$.
3. ATC at 20 units: \$10.
4. Profit $= (14 - 10) \times 20 = \$80$.



Source of the numbers: demand $P = 22 - 0.4Q$; TC $= 90 + 5Q + Q^2/40$, so MC $= 5 + Q/20$ and ATC $= 90/Q + 5 + Q/40$.

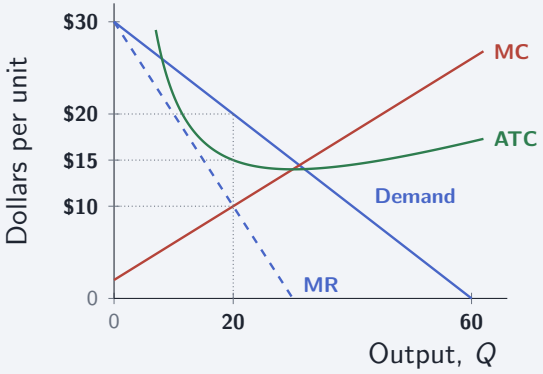
Part 2 summary

- In the short run the firm acts like a monopoly.
- Output: where MR = MC.
- Price: read from the demand curve at that output.
- Profit = $(P - \text{ATC}) \times Q$.
- If $P < \text{ATC}$, the firm makes a loss of $(\text{ATC} - P) \times Q$.

Your turn: Check your understanding

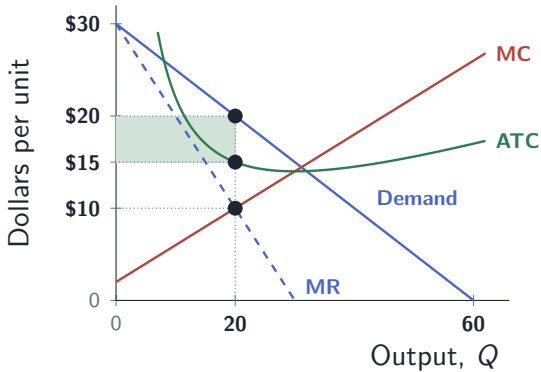
The firm maximizes profit.

1. Its output is:
   A. 10 B. 20 C. 60
2. Its price is:
   A. \$10 B. \$15 C. \$20
3. Its profit is:
   A. \$100 B. \$200 C. \$300



Solution: Check your understanding

1. **B.** MR meets MC at 20 units (both \$10).
2. **C.** Up to demand at 20 units: \$20.
3. **A.** ATC at 20 is \$15. Profit $= (20 - 15) \times 20 = \$100$.



Check: Part 2

In the short run, a firm operating in a monopolistically competitive market

A produces an output level where marginal revenue equals average total cost.

B produces an output where marginal revenue equals marginal cost, and the price is determined by demand.

C must earn zero economic profits.

D maximizes revenues as well as profits.

Check: Part 2 – answer

In the short run, a firm operating in a monopolistically competitive market

A produces an output level where marginal revenue equals average total cost.

B produces an output where marginal revenue equals marginal cost, and the price is determined by demand.

C must earn zero economic profits.

D maximizes revenues as well as profits.

Finding

Steps 1 and 2: output where MR = MC, then the price from the demand curve.

Part 3

The Long Run: Entry and Exit

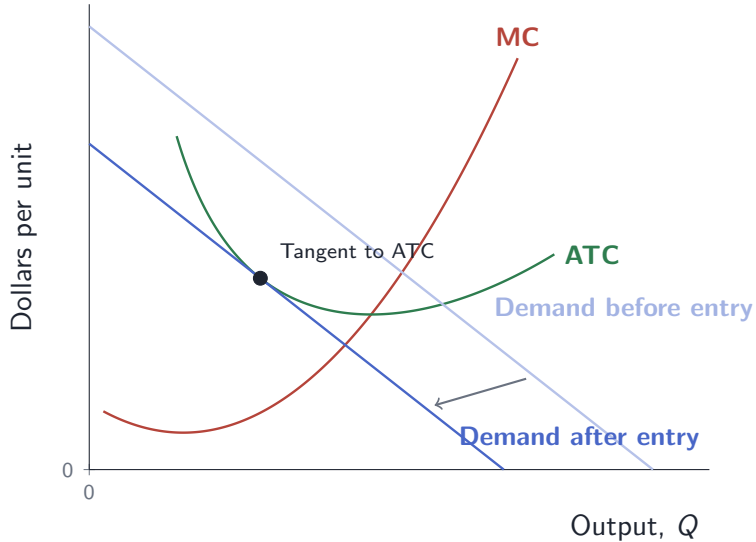
Profit attracts entry; losses cause exit

- **Profit:** new firms enter. Buyers now have more products to choose from.
- Each existing firm loses some customers: its demand curve shifts left.
- Entry continues until profit is zero.
- **Losses:** some firms exit. Their customers move to the firms that remain.
- Each remaining firm's demand curve shifts right, until the loss is zero.

Finding

Entry and exit move each firm's demand curve until profit is zero.

Entry on the graph



Long-run equilibrium: two conditions

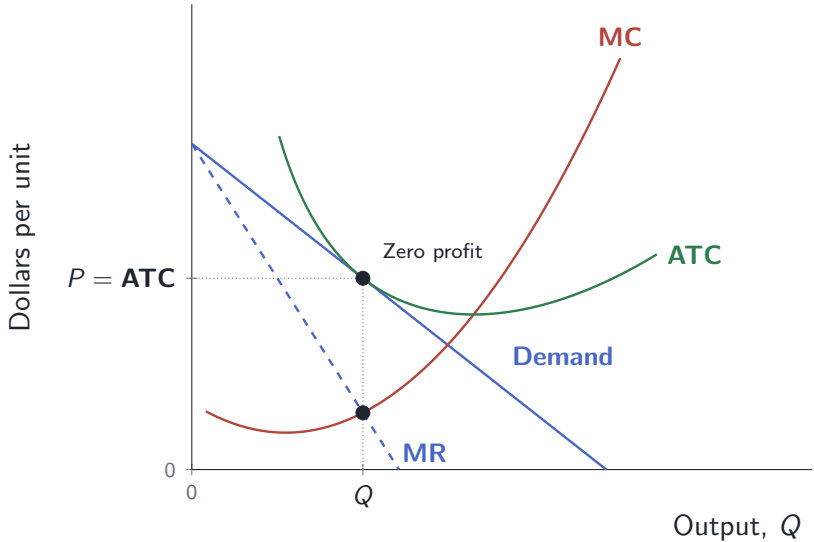
1. **The firm maximizes profit:** MR = MC.
2. **Profit is zero:** P = ATC.

- Profit = $(P - \text{ATC}) \times Q = 0 \times Q = 0$.
- On the graph, the demand curve just touches the ATC curve (it is tangent to ATC).

Finding

In the long run, MR = MC and P = ATC: demand is tangent to ATC and profit is zero.

Long-run equilibrium on the graph



Part 3 summary

- Profit attracts entry: each firm's demand shifts left.
- Losses cause exit: each remaining firm's demand shifts right.
- This stops when profit is zero.
- Long run: MR $=$ MC and $P =$ ATC.
- On the graph, demand is tangent to ATC.
- Compared with perfect competition: long-run profit is zero in both.

Your turn: Check your understanding

Coffee shops in a city are making a profit.

1. New shops open. Each existing shop's demand curve:
   A. shifts left. B. shifts right. C. does not move.
2. In the long run, each shop's profit is:
   A. positive. B. zero. C. negative.
3. In the long run, a shop sells 300 cups a day at \$4 a cup. Its ATC is:
   A. \$3. B. \$4. C. \$5.

Solution: Check your understanding

1. **A.** Buyers spread over more shops, so each shop sells less at every price.
2. **B.** Entry continues until profit is zero.
3. **B.** Zero profit means $P = \text{ATC}$, so ATC $= \$4$.

Check: Part 3

In monopolistically competitive markets, free entry and exit suggests that

A the market structure will eventually be characterized by perfect competition in the long run.

B all firms earn zero economic profits in the long run.

C some firms will be able to earn economic profits in the long run.

D some firms will be forced to incur economic losses in the long run.

Check: Part 3 – answer

In monopolistically competitive markets, free entry and exit suggests that

A the market structure will eventually be characterized by perfect competition in the long run.

B all firms earn zero economic profits in the long run.

C some firms will be able to earn economic profits in the long run.

D some firms will be forced to incur economic losses in the long run.

Finding

Entry removes profit and exit removes losses, so long-run profit is zero.

Part 4

Monopolistic Competition versus Perfect Competition

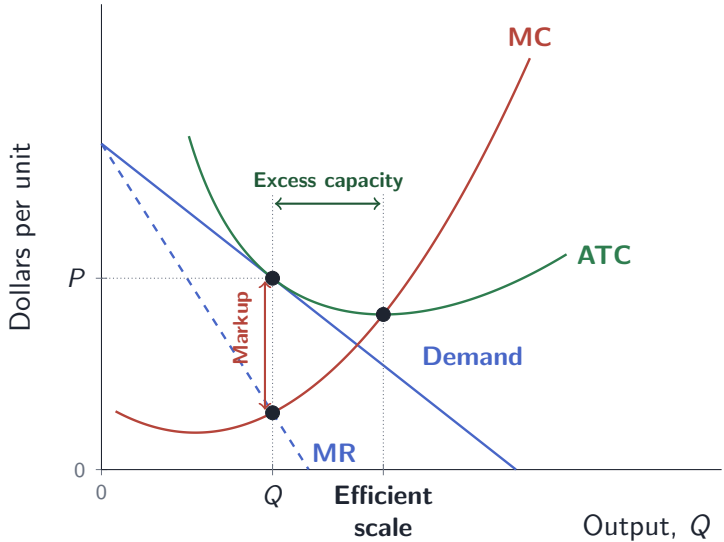
Two differences in the long run

Definition: Efficient scale

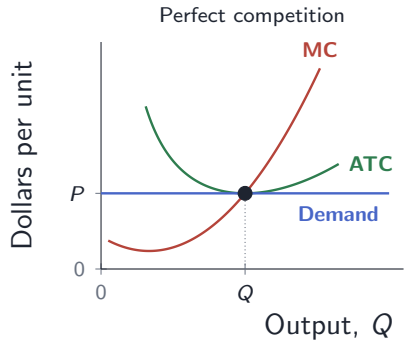
The output at which average total cost is at its minimum.

- A competitive firm produces at the efficient scale, with $P = \text{MC}$.
- **Excess capacity:** a monopolistically competitive firm produces *less* than the efficient scale.
- **Markup:** it charges a price above marginal cost, $P > \text{MC}$. The markup is $P - \text{MC}$.

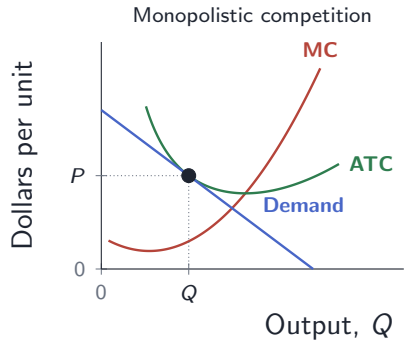
Excess capacity and markup on the graph



The two long runs side by side



$P = \text{MC}$. Output at the efficient scale.



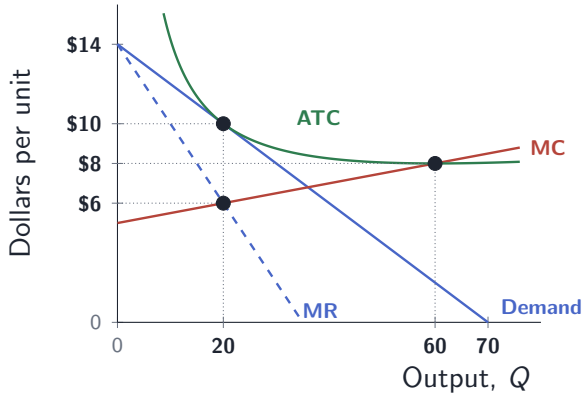
$P > \text{MC}$. Output below the efficient scale.

Finding

Profit is zero in both. Only perfect competition reaches the efficient scale with $P = \text{MC}$.

Example: a firm in long-run equilibrium

1. MR = MC at $Q = 20$; MC = \$6.
2. Demand at 20: $P = \$10 =$ ATC. Profit is zero.
3. Markup $= P - \text{MC} = 10 - 6 = \$4$.
4. Efficient scale: 60 units, ATC = \$8.
5. Excess capacity $= 60 - 20 = 40$ units.



Source of the numbers: demand $P = 14 - 0.2Q$; $\text{TC} = 90 + 5Q + Q^2/40$, so $\text{MC} = 5 + Q/20$ and $\text{ATC} = 90/Q + 5 + Q/40$.

The comparison in a table

Long run	Perfect competition	Monopolistic competition
Profit	Zero	Zero
Price and ATC	$P = \text{ATC}$	$P = \text{ATC}$
Price and MC	$P = \text{MC}$	$P > \text{MC}$: a markup
Output and efficient scale	At the efficient scale	Below it: excess capacity

Finding

Because $P > \text{MC}$, some buyers who value a unit above its cost do not buy it. That is why monopolistic competition is inefficient.

Part 4 summary

- Efficient scale: the output at minimum ATC.
- Excess capacity: the firm produces less than the efficient scale.
- Markup: the firm charges $P > \text{MC}$; the markup is $P - \text{MC}$.
- Long-run profit is zero, as in perfect competition.
- Compared with perfect competition: a competitive firm produces at the efficient scale with $P = \text{MC}$.

Your turn: Check your understanding

A firm in long-run equilibrium sells 20 units at \$15. Its MC at 20 units is \$10. Its ATC is lowest, \$14, at 30 units.

1. The markup is:
   A. \$1 B. \$5 C. \$15
2. The excess capacity is:
   A. 0 units B. 10 units C. 30 units
3. A competitive firm with these costs would, in the long run, charge:
   A. \$10 B. \$14 C. \$15

Solution: Check your understanding

1. **B.** Markup $= P - \text{MC} = 15 - 10 = \$5$.
2. **B.** Efficient scale minus output: $30 - 20 = 10$ units.
3. **B.** In the long run a competitive firm produces at the efficient scale, so $P =$ minimum ATC $= \$14$.

Check: Part 4

In a long-run equilibrium,

A only a perfectly competitive firm operates at its efficient scale.

B only a monopolistically competitive firm operates at its efficient scale.

C neither a competitive firm nor a monopolistically competitive firm charges a markup over marginal cost.

D both a perfectly competitive firm and a monopolistically competitive firm operate at their efficient scale of production.

Check: Part 4 – answer

In a long-run equilibrium,

A **only a perfectly competitive firm operates at its efficient scale.**

B only a monopolistically competitive firm operates at its efficient scale.

C neither a competitive firm nor a monopolistically competitive firm charges a markup over marginal cost.

D both a perfectly competitive firm and a monopolistically competitive firm operate at their efficient scale of production.

Finding

The monopolistically competitive firm has excess capacity: it produces below the efficient scale.

Which rule answers which question

Question	Rule
What defines the market?	Many firms, differentiated products, free entry.
What demand does a firm face?	Downward-sloping, so $P > \text{MR}$.
How much does it produce?	$\text{MR} = \text{MC}$.
What price does it charge?	The price on the demand curve at that output.
Short-run profit?	$(P - \text{ATC}) \times Q$; a loss if $P < \text{ATC}$.
Long-run profit?	Zero: $P = \text{ATC}$, demand tangent to ATC.
How does it compare with competition?	Excess capacity and a markup, $P > \text{MC}$.

Part 5

Advertising and Brand Names

Advertising: the critique and the defense

Firms with differentiated products advertise. The more differentiated the product, the more they advertise: cold medicines are advertised heavily on television; industrial drill presses are not.

The critique

- Advertising shapes tastes rather than informing.
- It makes demand less elastic.
- So the firm can charge a larger markup, $P - \text{MC}$.

The defense

- Advertising tells buyers about prices, new products and new sellers.
- That makes demand more elastic and competition stronger.
- So groups of sellers sometimes lobby to restrict advertising.

Example: advertising as a signal of quality

Two firms launch a new frozen pizza. A national advertising campaign costs \$3 million.

- **Good pizza:** buyers who try it come back. Profit from sales: \$5 million. With the campaign: $5 - 3 = +\$2$ million.
- **Poor pizza:** buyers try it once and do not return. Profit from sales: \$1 million. With the campaign: $1 - 3 = -\$2$ million.
- Only the firm with the good pizza advertises.

Finding

The willingness to spend heavily on advertising signals quality. What the advertisement says matters less than the fact that it is expensive.

Brand names

Example: brand-name ibuprofen and a store brand contain the same active ingredient, but the brand costs more.

The critique

- Buyers pay more for a brand that is the same as the generic.
- The brand creates a difference that is not real.

The defense

- A brand tells buyers about quality when quality is hard to judge: a traveler picks a hotel chain in a strange town.
- The firm has a reason to keep quality high, to protect its name.

Check: Part 5

If a firm in a monopolistically competitive market successfully uses advertising to decrease the elasticity of demand for its product, the firm will

A be able to increase its markup over marginal cost.

B eventually have to reduce price to remain competitive.

C increase the welfare of society.

D reduce its average total cost.

Check: Part 5 – answer

If a firm in a monopolistically competitive market successfully uses advertising to decrease the elasticity of demand for its product, the firm will

A be able to increase its markup over marginal cost.

B eventually have to reduce price to remain competitive.

C increase the welfare of society.

D reduce its average total cost.

Finding

This is the critique of advertising: less elastic demand lets the firm raise P further above MC.

Part 6

Measuring Market Concentration

How concentrated is a market?

- **Monopoly:** one firm has 100% of the market. Concentration is at its highest.
- **Perfect competition:** many small firms, each with a tiny share. Concentration is close to zero.
- Every other market structure lies between these two end points.
- Economists measure where a market lies with two common measures: the concentration ratio and the Herfindahl–Hirschman index (HHI).

The concentration ratio

Definition: Four-firm concentration ratio, CR4

The combined market share of the four largest firms in a market.

- **Market A:** four firms with shares 40%, 30%, 20% and 10%.
  $\text{CR4} = 40 + 30 + 20 + 10 = 100\%$.
- **Market B:** 20 firms with 5% each.
  $\text{CR4} = 5 + 5 + 5 + 5 = 20\%$.

Finding

A high CR4 means a few firms dominate the market. A low CR4 means sales are spread over many firms.

The Herfindahl–Hirschman index

Definition: Herfindahl–Hirschman index, HHI

The sum of the squared market shares (in percent) of all firms in the market.

- Monopoly: $\text{HHI} = 100^2 = 10{,}000$. Perfect competition: close to 0.
- **Market A** (40, 30, 20, 10): $1{,}600 + 900 + 400 + 100 = 3{,}000$.
- **Market C** (70, 10, 10, 10): $4{,}900 + 100 + 100 + 100 = 5{,}200$.
- Both have CR4 $= 100\%$. The HHI is higher for C: squaring gives the largest firm more weight.

U.S. antitrust agencies treat a market with an HHI above 1,800 as highly concentrated (Department of Justice and Federal Trade Commission, *Merger Guidelines*, 2023).

Two real markets

Industry (United States, 2017)	CR4
Breweries	68.6%
Furniture stores	21.5%

- Breweries: the four largest brewers sell more than two-thirds of the beer.
- Furniture stores: the four largest chains sell about one-fifth; many small stores sell the rest.

Source: U.S. Census Bureau, 2017 Economic Census, as reported in R. Kulick and A. Card, *Industrial Concentration in the United States: 2002–2017*, NERA Economic Consulting, 2022. Concentration has risen in over 75% of U.S. industries since the late 1990s: G. Grullon, Y. Larkin and R. Michaely, "Are US Industries Becoming More Concentrated?", *Review of Finance*, 2019; see also D. Autor et al., "The Fall of the Labor Share and the Rise of Superstar Firms", *Quarterly Journal of Economics*, 2020.

Check: Part 6

Suppose there is a market in which the firms hold the following market shares: 25%, 20%, 18%, 15%, 8%, 7%, 4%, 2%, 1%. What is the concentration ratio for this market?

Check: Part 6 – answer

Suppose there is a market in which the firms hold the following market shares: 25%, 20%, 18%, 15%, 8%, 7%, 4%, 2%, 1%. What is the concentration ratio for this market?

$$\text{CR4} = 25 + 20 + 18 + 15 = 78\%.$$

Finding

Add the shares of the four largest firms only. The other five firms do not count.